

Cognition-enhanced Meditation: A VR Mind-body Early Intervention for Adolescent Anxiety Disorders



Figure 1: An adolescent undergoes a VR meditation intervention session for anxiety disorders using the proposed CeM. (a) CeM comprehensively assesses the adolescent's historical HAMA-14, GAD-7, EMA, and MDQ scale data to determine the optimal stimulus and awareness intervention coefficients for the current session. (b) Based on the optimal stimulus coefficients, CeM guides the adolescent through structured Taichi movements using dynamic cues and the proposed Taichi-guided stimulus mode, rapidly inducing a cognition enhancement state. (c) Based on the optimal awareness coefficients, CeM utilizes awareness cues combined with personalized meditation audio expertly designed by an anxiety disorder therapist to conduct the meditation intervention, effectively alleviating the adolescent's anxiety level. (d) CeM quantitatively evaluates the intervention results from both the stimulus and awareness phases, yielding corresponding phase scores and a total score. These scores are used to positively reinforce the intervention's effectiveness, thereby enhancing the adolescent's intrinsic motivation to re-engage with the therapy.

ABSTRACT

Adolescent anxiety disorder has become an urgent issue in global public health. Although VR meditation therapy, as a non-pharmacological intervention, has shown great potential in alleviating anxiety symptoms in adolescents, existing methods have limitations. Current approaches overlook cognitive deficits in adolescents with anxiety disorders, such as attentional distraction and impaired executive function; intervention content is monotonous, lacks personalized design to accommodate individual adolescent differences; and it is difficult to achieve a natural transition between physiological arousal and psychological calm, highlighting the need to enhance the effectiveness of interventions for adolescent anxiety. To address these challenges, this paper proposes a cognition-enhanced meditation therapy (CeM). First, CeM constructs a mind-body early intervention framework that integrates structured physical exercise with VR meditation, leveraging exercise-induced cognition enhancement effects to improve adolescents' acceptance and integration into VR meditation. Second, CeM designs a condition-adaptive intervention control module that adaptively adjusts intervention content based on the adolescent's specific condition, improving adolescent adherence and ensuring long-term therapeutic effects. Furthermore, CeM introduces a Taichi-guided stimulus mode, utilizing Taichi's balance of movement and stillness to promote a seamless transition from an active state to a meditative mindset, enhancing cognitive preparation through adaptive movement arrangements and high user-experience movement guidance. Compared with existing state-of-the-art VR meditation therapies, CeM achieves a significant reduction in anxiety levels and related symptoms after eight intervention sessions over four weeks.

Index Terms: VR Therapy, anxiety disorder intervention, early intervention, VR meditation.

1 INTRODUCTION

According to World Health Organization statistics, 5.3% of adolescents experience anxiety disorders in 2025, constituting an urgent global public health issue [72]. Furthermore, adolescence is a critical risk period for the development of anxiety syndromes; without timely intervention, mild prodromal symptoms can easily progress into diagnosable anxiety disorders [7]. Therefore, early intervention for adolescent anxiety disorders is crucial, as intervening before mild psychological distress evolves into a lifelong mental illness halts disease progression and improves long-term prognosis.

Although traditional pharmacological treatments can alleviate early adolescent anxiety symptoms, they are accompanied by adverse effects such as sedation, impaired attention, mood fluctuations, and drug tolerance [4, 6, 33, 51]. Virtual Reality (VR) therapy, as a non-pharmacological intervention, has demonstrated great potential in the intervention of anxiety disorders due to its immersive experience and high intervention adherence [16, 22, 23]. Existing VR intervention therapies are mainly divided into three types. Firstly, exposure therapy simulates real-life anxiety-inducing scenarios, allowing patients to gradually adapt to anxiety triggers in a safe and controllable virtual environment, thereby reducing their sensitivity [3, 8, 43]. Secondly, behavioral activation therapy designs structured interactive tasks within VR scenes, combining real-time feedback and reward mechanisms to effectively promote positive behavioral engagement and reduce avoidance behaviors, which in turn improve emotion regulation capabilities [11, 15, 52]. Thirdly, meditation therapy constructs natural immersive environments and uses guided mindfulness exercises to cultivate patients' awareness and acceptance of present-moment experiences, thereby alleviating anxiety symptoms [20, 35, 36]. Among these, VR meditation therapy is particularly suitable for early intervention for adolescent anxiety disorders due to its non-invasive nature, low cognitive load requirements, and high environmental customizability, and has been proven by clinical studies to yield therapeutic effects [10, 76].

However, in the practical application of early intervention for adolescent anxiety disorders, VR meditation therapy exhibits mech-

anistic and technical limitations, primarily facing three major challenges. The first challenge is that existing therapies rely on background audio or virtual guides to direct meditation, but overlook the cognitive deficits present in adolescents with early-stage anxiety disorders, such as distractibility and impaired executive function, resulting in suboptimal meditation outcomes and limited intervention efficacy. The second challenge is that the current intervention content is monotonous and lacks personalized and continuous intervention designs tailored to the adolescents' specific conditions. This reduces patient compliance and makes it difficult to further improve long-term intervention outcomes. The third challenge is that existing therapies lack a natural transition mechanism between physiological arousal and psychological calmness, making it difficult to smoothly shift from an anxious state to a mental state suitable for meditation within a short period of time, thereby compromising the efficiency of individual intervention sessions.

In this paper, we propose a cognition-enhanced meditation therapy (CeM) to address the aforementioned challenges. Studies show that a single bout of exercise enhances executive function and attentional control by increasing blood flow and activation levels in the prefrontal cortex [28, 44], and improves cognitive functions by promoting the release of neurotransmitters [9]. Therefore, to address the first challenge, we construct a mind-body early intervention framework, which firstly integrates structured physical exercise with meditation, fully leveraging the exercise-induced cognition enhancement effects to improve the acceptance and integration of VR meditation. To address the second challenge, we propose a condition-adaptive intervention control module, which adaptively adjusts the intervention content and coefficients based on the adolescent's specific condition, thereby improving intervention compliance and efficacy. Furthermore, research shows that both immediately following and after a delay following light exercise, cognitive abilities such as attention and executive function show significant improvement [13]. Consequently, to address the third challenge, we introduce a Taichi-guided stimulus mode, which facilitates a seamless transition from activity to stillness by utilizing Taichi's characteristic movement-stillness interplay, and efficiently induces a sustained 'light exercise state' to enhance cognitive readiness through adaptive Taichi movement scheduling and high user-experience movement guidance, thereby potentiating the meditation intervention effects.

Figure 1 illustrates an adolescent undergoing a VR meditation intervention session for anxiety disorders using the proposed CeM. CeM integrates the stimulus and awareness intervention mechanisms through the constructed mind-body early intervention framework. Subsequently, CeM utilizes a condition-adaptive intervention control module to dynamically determine optimal intervention coefficients. CeM then guides the adolescent through efficient Taichi movements using the proposed Taichi-guided stimulus mode to enhance cognitive readiness for VR meditation, ultimately significantly improving the effectiveness of anxiety intervention for adolescents through VR meditation.

In summary, the contributions of this paper are as follows:

- A cognition-enhanced meditation therapy that achieves significant early intervention effects for adolescent anxiety disorders compared with state-of-the-art VR meditation therapy through comprehensive user studies;
- A mind-body early intervention framework that firstly integrates structured physical exercise with VR meditation to improve early intervention effects for adolescents with anxiety disorders;
- A condition-adaptive intervention control module that optimizes the intervention content and coefficients to promote intervention compliance and efficacy;
- A Taichi-guided stimulus mode to enhance cognitive functions through high user-experience movement guidance and

promote seamless transitions from stimulus to awareness.

The source code of this paper will be released.

2 RELATED WORK

In this section, we review the state-of-the-art research on VR therapy for mental disorders, and meditation-based applications for anxiety intervention that are related to our work.

2.1 VR Therapy for Mental Disorders

In recent years, VR therapy for mental disorders has advanced across three primary dimensions: integrating guided incentive therapies with biofeedback to improve intervention adherence; developing cloud-based collaborative environments to enhance the accessibility of remote digital therapeutics; and innovating interaction paradigms to deepen immersion and therapeutic engagement.

In research on biofeedback-guided therapies, researchers have utilized multisensory integration to optimize therapeutic outcomes. Combining VR with robotic exoskeletons [19] or specialized visual tasks [70] effectively motivates physical and visual rehabilitation. In mental health and addiction, realistic cue exposure paired with approach-avoidance training helps patients manage real-world triggers [45]. Furthermore, incorporating passive tactile feedback [66], real-time physiological monitoring, such as ECG and respiration [64], and dynamic environmental adjustments [5] significantly enhances bodily ownership and spatial presence during mindfulness meditation. For adaptive interventions, tracking gaze [39, 40] or utilizing brain-computer interfaces [55] enables dynamic tailoring of virtual environments based on patients' attention and neural feedback, thereby maximizing rehabilitation efficacy.

To overcome spatiotemporal barriers, cloud-based VR has been widely adopted to facilitate remote interventions. Collaborative VEs have been customized for remote art therapy and loving-kindness meditation [34], demonstrating positive effects on intimate relationships [68] and pain tolerance [31]. For clinical rehabilitation, cloud platforms tailored for spinal cord injuries enable remote data monitoring and optimize shared decision-making [27]. Researchers have established design guidelines for integrating collaborative VEs into daily mindfulness practices and addressing technical challenges in VR livestreaming [36, 37].

Regarding novel interaction paradigms, customized VR interfaces have been developed to enhance daily functioning and social skills across specific populations, including individuals with autism, agoraphobia, mild dementia, and mobility impairments [1, 58, 63, 71]. The design of virtual elements also profoundly impacts therapeutic efficacy; studies reveal that lower-fidelity humanoid avatars can offer advantages in interactivity [18], while virtual coaches with empathetic non-verbal cues significantly boost patient engagement [69]. Furthermore, recent innovations emphasize therapeutic customizability and emotional resonance, including parameterized VE construction for exposure therapy [21], interactive VR music generation [11, 12], role-swapping paradigms to foster empathy [24], and immersive group consultations to reduce clinical stigma [73]. In addition, fostering user autonomy in emotional expression and visualizing health data have emerged as effective strategies to augment emotional VR therapy [14, 65].

Although VR therapies have made significant progress for mental disorders, existing research primarily focuses on psychological disorders in adult populations, lacking early and precise intervention strategies tailored to adolescents with anxiety disorders. We propose a condition-adaptive intervention control module specifically designed for the neurocognitive characteristics and symptom manifestations of adolescents with anxiety disorders. By adaptively personalizing the intervention content, this module effectively achieves precise early intervention for adolescent anxiety disorders and significantly improves intervention compliance and efficacy.

2.2 Meditation-based Applications for Anxiety Intervention

Peterson et al. [53] systematically validated the effectiveness of mindfulness meditation for anxiety disorders for the first time, laying the groundwork for subsequent clinical applications and demonstrating that this method significantly reduces anxiety levels and helps maintain long-term symptom remission. Building upon this foundational work, recent research has focused on refining cognition-driven therapies to enhance intervention outcomes, reinforcing efficacy through multiple contextual strategies, and integrating multi-modal intelligent interactions to broaden clinical applicability and personalization.

To refine cognition-driven therapies, researchers have established structured neurocognitive frameworks to optimize intervention outcomes. Foundational models emphasize core dimensions, such as attentional regulation, emotional reappraisal, and self-acceptance, demonstrating that structured attention training improves executive and physiological functions [29, 42, 57]. Practically, flexible and simplified cognitive-behavioral protocols have proven superior to traditional lengthy sessions in promoting patient adherence [30]. Translating these cognitive principles into immersive environments, VR meditation has been shown to outperform conventional stress-relief methods [26]. Recent innovations construct AI-assisted VR mindfulness systems, which effectively leverage gamified and structured interactions to elevate mindfulness levels and alleviate anxiety among target populations [54].

Contextual and multisensory strategies have also been incorporated into standard meditation to reinforce therapeutic efficacy. Music-based interventions utilize rhythm synchronization and emotional reflection to modulate limbic responses, yielding significant improvements in social anxiety and psychological resilience [41, 75]. Similarly, parasympathetic activation through humor-driven contextual simulations effectively regulates stress hormones [2]. For highly specific, high-stress domains, e.g., competitive esports, integrating scenario-aligned imagery training with mindfulness familiarizes both body and mind with critical stressors, facilitating robust, real-time anxiety management [47].

Over the past five years, the integration of multi-modal intelligent interactions has significantly broadened the accessibility and precision of mindfulness therapies. Mobile applications such as Medito [74] and Insight Timer [48] offer guided meditations that ease anxiety, manage stress, and improve sleep, thus extending the reach and accessibility of mindfulness interventions for anxiety. Furthermore, advanced hardware has enabled deeper personalization. For instance, incorporating social robots into cognitive behavioral therapy provides realistic contextual simulations for treating social anxiety [56], while contactless millimeter-wave radar allows for continuous, unobtrusive stress quantification to inform personalized interventions [38]. Within the VR domain, structured platforms like ZenVR [20] synthesize these advancements by offering progressive, beginner-friendly meditation sessions augmented with real-time affective-behavioral feedback, thereby systematically fostering mindfulness skills.

Although existing VR meditation therapies alleviate anxiety symptoms, they overlook the inherent cognitive deficits of adolescents with anxiety disorders when applied to this group, such as attentional distraction and impaired executive function, thereby limiting the efficacy of meditation intervention. We propose a mind-body early intervention framework combined with a Taichi-guided stimulus mode, organically integrating structured physical exercise with meditation. This system leverages the cognition enhancement effect induced by exercise to efficiently guide adolescents into a cognitively enhanced state, and then administers adaptive meditation interventions within this state, thereby significantly improving the early intervention effects of meditation therapy.

3 COGNITION-ENHANCED MEDITATION

This section demonstrates the details of the proposed CeM. First, we introduce the mind-body early intervention framework in subsection 3.1. Then, we detail the condition-adaptive intervention control module in subsection 3.2, and describe the Taichi-guided stimulus mode in subsection 3.3.

3.1 Mind-body Early Intervention Framework

This mind-body early intervention framework integrates structured physical exercise into VR meditation therapy. This integration is designed to harness the cognition-enhancing effects of light physical activity [9, 28, 44], thereby mitigating the inherent limitations of purely static meditation and markedly improving the efficacy of early intervention for adolescent anxiety disorders. The subsequent subsections will first delineate the detailed therapy procedure of this framework, and then delve into the core virtual content required to support a single therapy session.

3.1.1 Therapy Procedure

The entire therapy course of CeM has eight sessions to achieve long-term intervention effects [32]. In each session, CeM integrates exercise and meditation guided by behavioral theory (BT) and cognitive behavioral therapy (CBT) to deliver effective mind-body therapy in VR. Thus, each session in the CeM therapy course has four phases: *intervention control*, *stimulus*, *awareness*, and *reinforcement*, as visualized in Figure 2.

In each intervention session, *intervention control* adaptively determines the intervention coefficients; *stimulus* and *awareness* guide adolescents in exercise stimulation and meditation awareness based on the intervention coefficients; *reinforcement* provides intervention quality scores to promote effective self-motivation and guides the next intervention session. After eight intervention sessions, the entire CeM therapy course is completed.

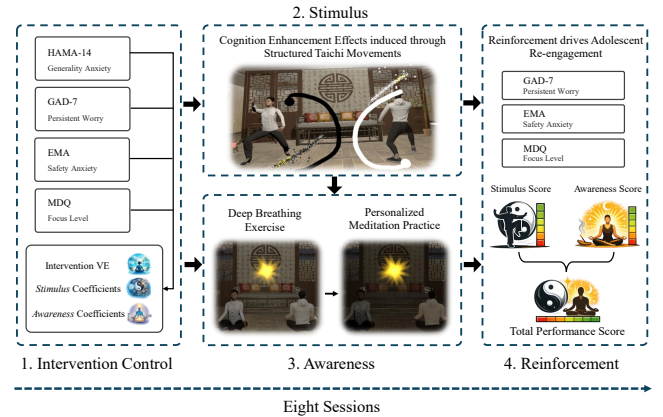


Figure 2: Visualization of the CeM Therapy Procedure

Intervention Control. In this phase, CeM determines the optimal intervention coefficients to maximize the effectiveness of the current session's intervention, including the intervention VE, the content and duration of exercise during *stimulus*, and the content and duration of meditation during *awareness*. To enhance the effectiveness and adherence of the intervention, we propose the condition-adaptive intervention control module in subsection 3.2, which adopts a personalized adaptive intervention control tailored to individual adolescent symptoms. This phase systematically assesses anxiety levels, exercise capabilities, and meditation focus in adolescents, and determines optimal intervention coefficients based on the assessment results.

Stimulus. Adolescents with anxiety disorders experience significant difficulties in attention control and psychological self-regulation. They struggle to focus on intervention guidance, are easily distracted, and lose motivation for intervention, making it challenging to maintain prolonged attention during the awareness

phase. In this phase, based on BT, CeM provides stimuli to adolescents through guided interactive physical exercise to enhance their cognition for the subsequent *awareness*. To increase the enjoyment of exercise and precisely guide exercise intensity to effectively enhance cognition, we propose a Taichi-guided stimulus mode in [subsection 3.3](#), which dynamically and precisely adjusts the duration and complexity of exercises through carefully designed interactive Taichi exercises based on preset intervention coefficients.

Awareness. In this phase, CeM provides tailored meditation for adolescents based on the CBT framework. Utilizing enhanced cognition gained from the *Stimulus* phase, this phase first guides adolescents into a meditative state through deep breathing exercises. Then, the designed guided audio verified by anxiety disorder therapists, leads adolescents through personalized, progressive meditation practices. The goals of this phase are to reduce the intensity and duration of adolescent anxiety, cultivate inner acceptance and calm, promote the improvement of emotional regulation skills, and enhance psychological resilience. By observing their inner conscious experiences, individuals can be assisted in re-establishing connection with their own mental and physiological states, thereby promoting self-regulation and anxiety alleviation.

Reinforcement. The intervention outcomes need to be promptly reinforced, as this determines adolescents' willingness to re-engage. After adolescents complete the *awareness* phase, we guide them to complete the GAD-7, EMA, and MDQ scales, thereby reinforcing their perception and awareness of improvements in their anxiety levels. Then, CeM generates the *stimulus* score η_{sti-m} , the *awareness* score η_{mdq} , and the intervention quality score η_{int} . The three scores are visually presented as positive feedback for reinforcement at the end of each session. It aims to enhance adolescents' motivation to participate, boost their self-confidence, and thus promote sustained positive experiences. A higher score indicates greater engagement. CeM derives the *stimulus* score η_{sti-m} by quantifying medium-difficulty exercise performance (as shown in [Equation 8](#)), and collects statistical data from the MDQ scale (see details in the supplementary material subsection 1.4) to obtain the *awareness* score η_{mdq} . Furthermore, η_{int} is the average of η_{sti-m} and η_{mdq} . η_{sti-m} and η_{mdq} scores both range from 0 to 1, and they provide personalized guidance for subsequent interventions, enhancing the effectiveness of interventions in *stimulus* and *awareness* phases. To avoid the potential psychological stress caused by low scores, all scores in the reinforcement phase are normalized to a high-score range of 80-100 before being presented to the adolescent.

3.1.2 Virtual Content

CeM summarizes the VR content design for adolescent anxiety interventions, covering three aspects: intervention VE types, stimulus interactions, and dynamic cues.

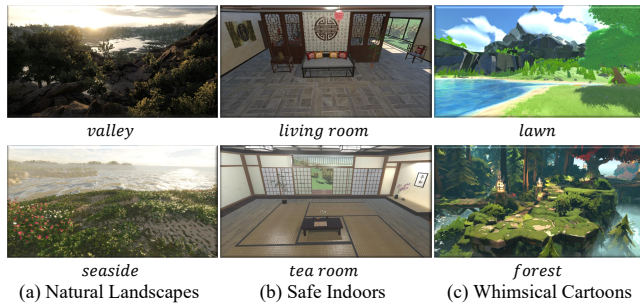


Figure 3: Visualization of Intervention VEs in CeM

Intervention VE Types. Research shows that the impact of different VEs on emotional regulation varies significantly [62]. Generalized anxiety manifests as persistent worry, a lack of security, and difficulty concentrating [60]. Therefore, we meticulously designed three intervention VE types: (a) natural landscapes, (b) safe

indoors, and (c) whimsical cartoons. These three types aim to simulate diverse environmental stimuli, each focusing on enhancing the alleviation of intense worry, a sense of security, and attention focusing. The natural landscape is primarily designed for anxious individuals with intense worry, providing a relaxing experience in a natural environment. The safe indoors offers a strong sense of security and a controlled environment to help alleviate anxiety. Cartoon scenes attract adolescents by increasing the element of fun. As shown in [Figure 3](#), each VE type contains two scenes: natural landscapes include *valley* and *seaside*, safe indoors include *living room* and *tea room*, whimsical cartoons include *lawn* and *forest* scenes.

Positive Psychological Hints. To enhance the engagement of interventions and improve adolescents' compliance, we employ carefully designed prompts to provide positive and gentle psychological feedback to adolescents to help them build positive emotional cognition. As shown in [Figure 4](#), when adolescents complete standard exercises according to the guidance, they receive praise, as shown in [Figure 4 \(a\)](#); if the movements are not performed correctly, encouraging prompts help guide them, as shown in [Figure 4 \(b\)](#). Additionally, if the adolescent's movement speed is too slow or too fast, we provide normative guidance through prompts, as shown in [Figure 4 \(c\)](#) and [\(d\)](#). We ensure that these interactive elements are simple and intuitive, making them easy for adolescents to understand and execute, thereby helping them better master the intervention treatment process.

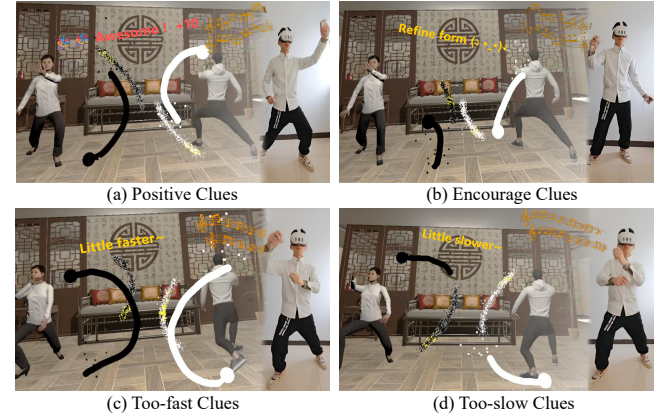


Figure 4: Visualization of Positive Psychological Hints in CeM

Dynamic Cues. To enhance the immersion and intervention experience, we construct a hierarchical dynamic cue set, as shown in [Figure 5](#), including temporal cues, stimulus cues, immersive cues, and awareness cues:

- Temporal cues indicate the duration and remaining time of the current intervention. We build an incense holder that measures time dynamically through the burning of two incense sticks, which correspond to the stimulus and awareness phases.
- Stimulus cues include a virtual mentor and sound prompts for the stimulus phase. The virtual mentor demonstrates the correct movements to provide guidance for adolescents to complete the exercises correctly, while clearly prompting them to carry out the current task, ensuring the smooth progression of stimulus.
- Immersive cues include auditory white noise and dynamic lighting that are consistent with visual semantics, enhancing adolescents' immersion and promoting deep relaxation. For example, in the VE *mountaintop*, we add sounds of leaves rustling in the wind and birds singing to enhance the immersion. The dynamic lighting transitions naturally from bright to dim, helping adolescents gradually calm down and regulate their emotional rhythm.
- Awareness cues include a breathing ball and mindfulness guidance audio. The breathing ball guides adolescents in deep breathing through changes in shape and size, helping them quickly enter

a meditative state. The mindfulness guidance audio, inspired by the concept of ZenVR, features personalized audio guidance for eight interventions, gradually leading adolescents to master and practice meditation.

These dynamic cues are interwoven to create a comprehensive therapy context aimed at enhancing adolescents' sense of involvement and the effectiveness of the intervention.

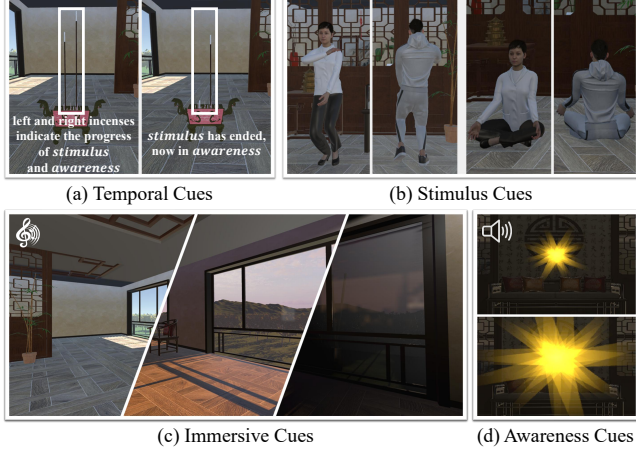


Figure 5: Visualization of Dynamic Cues in CeM

3.2 Condition-adaptive Intervention Control

Algorithm 1: Condition-adaptive Intervention Control

Input : intervention session i , intervention VE set VE , last-session VE ve_{i-1} , Taichi movements set $TAICHI$, meditation auditory set AUD , primary HAMA-14 η_{hama}^0 , last-session GAD-7 η_{gad}^{i-1} , last-session EMA η_{ema}^{i-1} , last-session MDQ η_{mdq}^{i-1} , last-session stimulus scores $[\eta_{sti-m}^{i-1}, \eta_{sti-c}^{i-1}]$

Output: this-session VE ve_i , $stimulus(Ts_i, TC_i)$, $awareness(AU_i, Ta_i)$

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// step 1: optimal intervention VE selection
1  $P_w \leftarrow \text{calProb}(\eta_{hama}^0, \eta_{gad}^{i-1}, \eta_{ema}^{i-1}, \eta_{mdq}^{i-1})$ 
2  $VE \leftarrow \text{initWeight}(VE)$ 
3  $ve_i \leftarrow \text{select}(VE, ve_{i-1}, P_w)$ 
// step 2: stimulus phase coefficients determination
4  $TC_l, TC_m, TC_h \leftarrow \text{decompose}(TAICHI)$ 
5  $TC_i \leftarrow \text{compose}(TC_l, TC_m, TC_h, \eta_{sti-m}^{i-1}, \eta_{sti-c}^{i-1})$ 
6  $Ts_i \leftarrow \text{calTime}(TC_i)$ 
// step 3: awareness phase coefficients determination
7  $AU_w, AU_s, AU_f, AU_{trans} \leftarrow \text{decompose}(AUD)$ 
8  $AU_i \leftarrow \text{compose}(AU_w, AU_s, AU_f, AU_{trans}, P_w)$ 
9  $Ta_i \leftarrow \text{duration}(AU_i)$ 
10 return  $ve_i, stimulus(Ts_i, TC_i), awareness(AU_i, Ta_i)$ 

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Conventional VR therapies for adolescent anxiety intervention are limited to predefined VEs with fixed intervention procedures. Adolescents with anxiety disorders are resistant to repetitive interventions. The immutable intervention reduces treatment adherence. Moreover, fixed procedures neglect the current anxiety state and prior intervention effects, making it challenging to customize personalized intervention plans based on individual differences, which ultimately decreases intervention effectiveness. To provide effective intervention measures for different adolescents, we propose a condition-adaptive intervention control module that offers personalized anxiety disorder intervention for each session throughout the entire therapy course.

Algorithm 1 outlines the procedure employed by the condition-adaptive intervention control module, which includes three steps: the optimal intervention VE selection (lines 1–3), the stimulus phase coefficients determination (lines 4–6), and the awareness phase coefficients determination (lines 7–10). Given the current intervention session i , the intervention VE set VE , the last-session VE ve_{i-1} , the Taichi movements set $TAICHI$, the meditation auditory set AUD , the primary HAMA-14 scale result η_{hama}^0 , the last-session GAD-7 scale result η_{gad}^{i-1} , the last-session EMA scale result η_{ema}^{i-1} , the last-session MDQ scale result η_{mdq}^{i-1} , and the last-session stimulus scores $[\eta_{sti-m}^{i-1}, \eta_{sti-c}^{i-1}]$, it outputs the optimal VE ve_i for this-session intervention, the duration Ts_i and exercise content TC_i to determine this-session stimulus phase, and the duration Ta_i and auditory content AU_i to determine this-session awareness phase.

We first assess the anxiety level in terms of generality, worry, safety, and focus as the guiding basis [60] to generate the personalized intervention (line 1). In terms of general anxiety level, we use the Hamilton anxiety rating scale (HAMA-14) [25] as the observer-rated anxiety level scale. In terms of persistent worry-based anxiety level, we use the generalized anxiety disorder scale (GAD-7) [59] as the self-reported anxiety level scale. We design the ecological momentary assessment scale (EMA) to refine the assessment of adolescents' self-reported anxiety levels in terms of safety. To assess adolescents' anxiety-based focus during intervention treatments, we design the Meditation Depth Questionnaire (MDQ) to quantify the anxiety-based focus level during meditation. In HAMA-14, GAD-7, and EMA, higher scores indicate higher levels of anxiety. Higher MDQ scores indicate less meditation concentration. For the details of HAMA-14, GAD-7, EMA, and MDQ scales, please refer to the supplementary material section 1.

Given that HAMA-14 is a clinician-rated scale requiring one-on-one administration by a therapist and is time-consuming (15–20 minutes per assessment), CeM utilizes it to assess the general anxiety levels of adolescents at the beginning and end of the entire therapy course. To avoid assessment fatigue and potential measurement bias caused by frequent, lengthy evaluations, we adopt a more efficient combination of self-report questionnaires (GAD-7, EMA, MDQ) after each individual session. With an average completion time of only 3–5 minutes, this combination allows for immediate, continuous, and dynamic monitoring of adolescents' anxiety levels and psychological states across three dimensions, persistent worry, sense of security, and focus, following each treatment session.

In line 1, we calculate the anxiety level assessment set $P_{adp}=[prop_w, prop_s, prop_f]$. Based on the scale results, we calculate the anxiety levels for persistent worry $prop_w$, security lacking $prop_s$, and focus deficit $prop_f$, as shown in Equation 1:

$$\begin{cases} prop_w = \frac{1}{i+1} |\eta_{hama}^0| + \frac{i}{i+1} |\eta_{gad}^{i-1}| \\ prop_s = \frac{1}{i+1} |\eta_{hama}^0| + \frac{i}{i+1} |\eta_{ema}^{i-1}| \\ prop_f = \frac{1}{i+1} |\eta_{hama}^0| + \frac{i}{i+1} |\eta_{mdq}^{i-1}| \end{cases} \quad (1)$$

where $|\cdot|$ is a regularization function used to normalize all scale results to the same range $[0, 1]$.

Next, we initialize the weight of each ve in VE to quantify intervention effects (line 2). Since generalized anxiety disorder manifests as persistent worry, lack of security, and difficulty concentrating, CeM constructs three VE types: natural landscapes NL , safe indoors SI , and whimsical cartoons WC , focusing on enhancing worry relief, safety feelings, and concentration, respectively. We define three weights to quantify the intervention effects for each ve : worry relief weight w_w , security enhancement weight w_s , concentration improvement weight w_f , all range from $[0, 1]$.

Benefiting from the superior capabilities of multi-modal large

language models (MLLMs) in visual perception and semantic reasoning, for each VE ve , CeM uses the MLLM GPT-5.2-Thinking [67] to obtain the quantification results of scene semantics and spatial attributes, and obtains the final three-dimensional intervention weights $[w_w, w_s, w_f]$ of ve , as detailed in supplementary material subsection 2.1. Then, we calculate the overall weight $ve.prop$ for each ve in VE using Equation 2:

$$ve.prop = ve.w_w \cdot prop_w + ve.w_s \cdot prop_s + ve.w_f \cdot prop_f \quad (2)$$

We select ve with the highest overall weight $ve.prop$ from VE as the intervention VE for this session, and to avoid repetition, we exclude the intervention scene from the last session (line 3).

After that, we perform the stimulus coefficients determination, which determines the duration Ts_i and exercise content TC_i in the stimulus phase (lines 4-6). Research shows that both immediately following and after a delay following light exercise, cognitive abilities such as attention and executive function show significant improvement [13]. Therefore, during the stimulus phase, adolescents are required to engage in light exercise for a duration of 8 minutes to enhance the effectiveness of subsequent awareness meditation. In the stimulus phase, the exercise should neither be too simple, which might make adolescents feel bored, nor too complex and exhausting, which would make it difficult for them to execute. In addition, it should effectively guide adolescents into the light exercise state. The natural transition characteristics of motion and stillness in Taichi allow it to seamlessly integrate with mindfulness therapy. Thanks to the ability of Taichi to efficiently guide light-state exercise and its natural transition between movement and stillness, it is suitable for CeM. In the stimulus phase, we use 24-form Taichi to guide adolescents in exercise. Since many complex movements in 24-form Taichi are difficult for adolescent participants, resulting in low exercise efficiency, we decompose the 24 forms into simple, medium, and complex difficulty movements (line 4), as detailed in supplementary material subsection 2.2. By dynamically combining movements of varying difficulty, CeM can more efficiently guide adolescents into light exercise states.

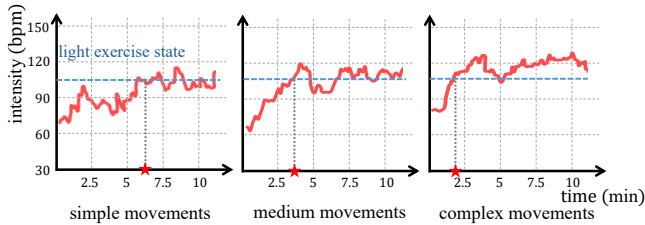


Figure 6: Time cost for Entering Light State

We recruit eight adolescent participants to perform Taichi exercises of varying difficulty levels, Figure 6 shows the time required for eight adolescent participants to enter light exercise states using Taichi at different difficulty levels. The light exercise state entry confirmation is calculated using the ACSM standard guideline [50] based on the maximum heart rate [61]. We find that as the difficulty level increases, the time taken to enter the exercise state is shortened. In other words, the higher the complexity of the movements, the greater the efficiency of the movement. In addition, we conduct a simple survey of the eight participants, and according to their feedback, they prefer performing Taichi movements of medium (62.5%) and simple difficulty levels (25.0%), rather than complex-difficulty movements (12.5%). Given the overall energy E that reaches a light exercise state and remains for 8 minutes, according to Figure 6, the intensity of Taichi movements in TC_l , TC_m , and TC_h per unit time is as Equation 3:

$$\Delta E_l = \frac{E}{\Delta T_l}, \Delta E_m = \frac{E}{\Delta T_m}, \Delta E_h = \frac{E}{\Delta T_h} \quad (3)$$

where ΔE_l , ΔE_m , and ΔE_h are unit exercise intensity in TC_l , TC_m , and TC_h , while ΔT_l , ΔT_m , ΔT_h refer to the duration to enter the light

exercise state and to maintain that state for eight minutes. Since adolescents prefer simple and medium difficulty movements over highly complex techniques, to maximize exercise efficiency based on user proficiency, we have Equation 4:

$$\begin{aligned} & \text{minimize } x \\ & \begin{cases} \Delta E_l \cdot p_l \cdot x + \Delta E_m \cdot p_m \cdot x + \Delta E_h \cdot p_h \cdot x \geq E \\ p_l = 1 - p_m - p_h \\ p_m = \frac{\eta_{sti-m}^{i-1}}{\eta_{sti-m}^{i-1} + \eta_{sti-c}^{i-1}} \\ p_h = (1 - p_m) \cdot \frac{\eta_{sti-m}^{i-1}}{\eta_{sti-m}^{i-1} + \eta_{sti-c}^{i-1}} \end{cases} \end{aligned} \quad (4)$$

Then, we select $p_l \cdot x$, $p_m \cdot x$, and $p_h \cdot x$ moves from TC_l , TC_m , and TC_h respectively to form the current session's Taichi movement set TC_i (line 5). The time duration of this-session stimulus phase Ts_i is the time cost of all movements in TC_i (line 6).

Then, we perform the *awareness* coefficients determination, which determines the duration Ta_i and auditory content AU_i in *awareness* (lines 7-9). To mitigate habituation arising from repetitive stimuli and achieve audio generation that balances therapeutic specificity with auditory coherence, we construct a diverse, hierarchically structured therapist-verified audio corpus, denoted as AUD (line 7). This corpus not only comprises intervention subsets targeting three dimensions—persistent worry (AU_w), lack of security (AU_s), and lack of focus (AU_f), but also incorporates a structured transition subset, AU_{trans} , which covers guidance for grounding, semantic transitions, and concluding awakening.

In line 8, we synthesize the audio AU_i for the current session. Initially, adhering to an exclusionary principle, core audio segments used in the preceding session AU_{i-1} are marked as temporarily deactivated, thereby removing them from the current candidate pool. Subsequently, the system retrieves and selects audio segments that precisely match the current therapeutic phase or emotional state from the remaining active audio library, based on a predefined probability value P_w . Following this, appropriate transitional phrases from AU_{trans} are inserted between the selected core audio segments to ensure a seamless and coherent audio flow. Finally, after professional confirmation by the therapist, the definitive audio AU_i for the current session is generated and finalized.

Then, in line 9, we set the total duration of the meditation audio Ta_i based on the duration of AU_i . Finally, this module outputs the determined VE ve_i for the current session, the coefficients required for the stimulus phase (Ts_i and TC_i), and the coefficients required for the *awareness* phase (AU_i and Ta_i) (line 10).

3.3 Taichi-guided Stimulus Mode

Cognitive function and attention control are enhanced immediately following a single session of acute exercise [13]. These effects benefit the intervention of mindfulness therapy interventions for adolescent anxiety. The natural transition characteristics of motion and stillness in Taichi allow it to seamlessly integrate with mindfulness therapy. However, traditional VR interactive Taichi methods, such as guiding through line drawings [49], evaluate the accuracy and standardization of Taichi movements by detecting collisions between designated guide lines and the user's motion trajectory. This design paradigm inherently restricts adolescents' freedom of movement, consequently diminishing their kinesthetic immersion and overall experiential quality within the VR environment. In addition, unrestricted movement makes it difficult to effectively guide adolescents in standard Taichi practice, thereby reducing exercise quality. To overcome the above limitation, we propose the Taichi-guided stimulus mode, which effectively instructs adolescents in performing standardized Taichi movements while maximizing the freedom of movement, ultimately improving exercise quality.

Benefiting from the intuitive visual guidance features of the dynamic flow in a particle system, in Taichi-guided stimulus mode, we use trajectory particles and correction particles to guide users to perform natural Taichi movements during the stimulus phase, as visualized in Figure 7. Trajectory particles are standardized virtual mentor Taichi two-hand motion standard trajectories, dynamically mapped based on the user's pose, and presented in the user's first-person virtual perspective. Correction particles provide real-time visual feedback on how the user's current hand movements are correcting toward the standard trajectories.

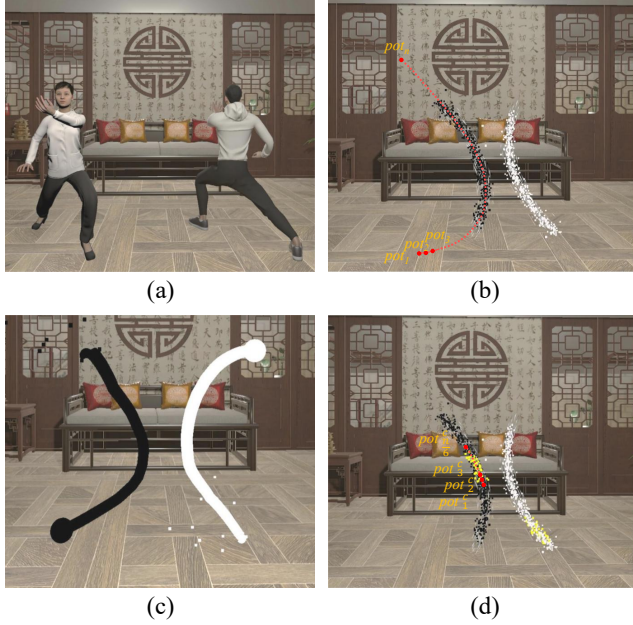


Figure 7: Visualization process of the trajectory and correction particles in Taichi-guided stimulus mode.

Figure 7 (a) visualizes the Taichi movements currently demonstrated by the virtual instructors. During the motion demonstration, we sample the position coordinates of a virtual instructor's left and right hands in VE at fixed Δt time intervals, thereby generating for each Taichi movement a trajectory sequence consisting of N ordered 3D points, named as the original standard trajectory point set $POINT'$. The value of N is determined by the duration of the Taichi movement demonstration. In the implementation, Δt is set to 200ms. To comprehensively characterize the dynamic characteristics of the movement, we assign an instantaneous velocity attribute $pot'_i.vel$ for each point pot'_i in $POINT'$, as shown in Equation 5:

$$pot'_i.vel = \begin{cases} \frac{|pot'_i.pos - pot'_{i-1}.pos|}{\Delta t}, & i \in [1, N) \\ 0, & i = 0 \end{cases} \quad (5)$$

where $pot'.pos$ is the sampled position coordinates of the point pot' . We use $POINT'$ to represent the instructor's standard motion trajectory of the current movement.

To better guide users of different heights and limb proportions to perform natural Taichi movements, we apply a normalization process with personalized sizing and spatial alignment based on the user's own head-hand pose, ensuring that the amplitude of the virtual trajectory matches the user's body proportions, thereby providing ergonomically comfortable guidance and avoiding discomfort or unnatural ranges of motion. Specifically, we assume that the distance from the chest to the head and the distance from the chest to the hands maintain a nearly constant proportional relationship. We compute the normalized standard trajectory point set $POINT$ of the normalized guiding trajectory. For each point pot_i in $POINT$, its

instantaneous velocity $pot_i.vel$ is the same as $pot'_i.vel$ in $POINT'$, and its position $pot_i.pos$ is obtained by Equation 6:

$$pot_i.pos = \frac{(pot'_i.pos - \overline{pot'.pos}) \cdot S_{user}}{S_{ref}} \quad (6)$$

where $pot'_i.pos$ is the sampled position coordinates of the corresponding point in $POINT'$, $\overline{pot'.pos}$ is the average of the sampled position coordinates of all points in $POINT'$, S_{ref} is the effective reach distance of the virtual instructor's hands relative to the head when both arms are fully extended, and S_{user} is the distance of the user's controllers relative to the HMD in the same fully extended arms posture. As shown in Figure 7 (b), we guide the trajectory particles generation in VE using all the points pot in $POINT$, and present it from the user's first-person perspective.

Figure 7 (c) visualizes the trajectories of the user's both-hand movements performing Taichi movements in real time while following the trajectory particles, which are depicted as continuous lines that allow the user to immediately observe their own movement path. The user's movement trajectories are still sampled at fixed time intervals Δt . To filter out minor tremors or brief pauses that may occur during the motion, we introduce an adaptive sampling strategy. Specifically, if the spatial distance between the current sampling point and the last valid collected point falls below a preset threshold, i.e., μ times the average distance between corresponding adjacent points in $POINT$, the user is considered to be in a relatively static or unintended pause state, and the sampling point is discarded. In the implementation, μ is set to 0.1. Only when the user resumes normal movement, i.e., when the distance between two points exceeds this threshold again, is a new trajectory point collected and added to the motion sequence. The resulting real-time set of user trajectory points is denoted as the user trajectory sample point set $POINT_u$. For any sampling point in $POINT_u$, its instantaneous velocity is calculated using Equation 5.

Figure 7 (d) visualizes the correction particles based on the correction sample point set $POINT_c$ for the user's current movements. These corrective particles are designed to guide the user's immediate future movement direction and speed, steering them toward the standard trajectory. Specifically, we use the last sampled trajectory points number M from $POINT_u$ as a temporal reference to dynamically extract $POINT_c$ from $POINT$. Centered around $POINT[M]$, $POINT_c$ extends forward and backward by $N/12$ standard trajectory points each (if available), forming a set comprising a total of $N/6$ points. To better guide the user's movement speed, for each point pot_i^c in $POINT_c$, its velocity $pot_i^c.vel$ is calculated by Equation 7:

$$pot_i^c.vel = \begin{cases} 2pot_i.vel - pot_i^u.vel, & i < M \\ pot_i.vel + pot_{N/12}.vel + pot_{N/12}.vel, & i \geq M \end{cases} \quad (7)$$

where pot_i is the i -th sampled point in $POINT$, and pot_i^u is the i -th point in $POINT_u$. Through this adaptive speed adjustment mechanism, when detecting that the user's movement speed is slower than the standard, corrective particles flow at a faster rate to intuitively prompt the user to accelerate. Conversely, if the user moves too quickly, the speed of the corrective particles decreases accordingly, guiding the user to slow down, thereby achieving precise and real-time movement speed correction.

In each Taichi movement, the user starts and ends the current movement by pressing and releasing the trigger. To avoid interference from accidental misoperations during data collection, if the number of valid user trajectory points captured by the system upon trigger release falls below the 10% threshold of the total standard trajectory points for that movement, the session is deemed invalid. The system continues waiting for a new valid movement initiation command to ensure data quality. Once the user completes the current movement, we generate the stimulus scores η_{sti-m} and η_{sti-c} to quantify their performance on medium- and complex-difficulty

movements to guide the next-session intervention. The scores η_{sti-m} and η_{sti-c} are both calculated using Equation 8 based on the standard accuracy of movement posture and the standard rhythm of speed in varying-difficulty movements:

$$\eta_{sti} = \sum_{i \in N} \{DTW(POINT_u[i].pos, POINT[i].pos) + DTW(POINT_u[i].vel, POINT[i].vel)\} / 2N \quad (8)$$

where DTW is the Dynamic Time Warping [46] based similarity calculation function with the value range normalized to [0,1].

4 MAIN USER STUDY

This main user study evaluates the clinical early intervention effects of CeM on adolescents with anxiety disorders within the meditation framework. To independently verify the advantages of the proposed Taichi-guided stimulus mode, we provide a pilot study in supplementary material section 3. The experimental results of the pilot user study indicate that the proposed mode significantly outperforms existing Taichi-guided baseline methods in terms of stimulus accuracy and user experience. All studies are approved by the IRB.

We propose two hypotheses for the main user study:

H1: After completing the eight-session intervention, compared to existing meditation therapies, adolescents with anxiety disorders that use *CeM* achieve significant intervention advantages in reducing anxiety;

H2: After completing the eight-session intervention, compared to existing meditation therapies, adolescents with anxiety disorders that use *CeM* achieve significant intervention advantages in alleviating adolescent anxiety disorder-related persistent worry, insecurity, and difficulty concentrating.

4.1 User Study Design

Apparatus. Our system uses a Quest 3 HMD powered by a workstation with a 3.8GHz Intel(R) Core(TM) i7-10700KF CPU, 32GB of RAM, and an NVIDIA GeForce GTX 3080Ti graphics card. The system runs at 90fps for each eye. Our program is developed with C# and HLSL, and runs in Unity 2021.3.8f1. To prevent the spread of germs through shared devices, we use alcohol wipes to disinfect the HMD before and after each use.

Participants. We recruited 24 adolescents with anxiety disorders as participants for this study based on the following inclusion and exclusion criteria. The inclusion criteria are as follows:

- An adolescent has been diagnosed with an anxiety disorder that meets the diagnostic criteria for anxiety disorders in DSM-5 [17];
- An adolescent is between 12 and 18 years of age, has full use of their hands and feet, possesses normal vision, hearing, and mobility, and can use VR HMDs;
- An adolescent voluntarily participates in the study, and their legal representative or guardian is informed and consents.

The exclusion criteria are as follows:

- An adolescent has been diagnosed with conditions such as Down syndrome, cerebral palsy, epilepsy, schizophrenia, bipolar disorder, or claustrophobia;
- An adolescent is currently participating in or has participated in related studies within the past 30 days.

Place and Time. The study lasts 4 weeks, with 2 sessions per week, each approximately 40 minutes, and was conducted in the psychiatric intervention department of a neurology hospital. In addition to the mindfulness meditation intervention in this study, all adolescents receive the same routine treatment provided by the department, 2 times per week for 30 minutes per session, consisting of repetitive transcranial magnetic stimulation (rTMS) and supportive psychological counseling provided in a fixed format.

Conditions. This study employs three different meditation therapies as interventions for adolescents with anxiety disorders: the

proposed *CeM*, the state-of-the-art clinical VR meditation system *ZenVR* [20], and conventional VR meditation therapy *mind*. *mind* is a VR meditation therapy conducted in *living room*. The meditation audios used in *ZenVR* and *mind* are consistent with *CeM*. In this study, *CeM* is referred to as *EG*, while *ZenVR* and *mind* are referred to as *CG1* and *CG2*.

Metrics. This study assesses participants' observer-rated anxiety levels using the HAMA-14 scale administered by therapists before the start and the end of the entire intervention course. After each intervention session, participants complete self-rated GAD-7, EMA, and MDQ scales to evaluate anxiety levels based on aspects such as persistent worry, sense of security, and concentration.

Task and Procedure. The experiment has three stages:

- **Pre-test.** First, we collect information from therapists for all participants, including name, gender, and age. With the therapists' assistance, participants then complete a pre-test to obtain their pre-intervention scores on the HAMA-14 scale. Concurrently, participants independently complete the GAD-7 and EMA self-assessment scales to comprehensively evaluate their anxiety and emotional states. To avoid potential confounding effects and ensure the accuracy of measurement results, the MDQ scale, primarily used to assess adolescents' concentration during meditation, is not administered before the start of the intervention. Participants are then assigned to the experimental group (*EG*) and the control groups 1 (*CG1*) and 2 (*CG2*) based on gender and age, with 8 participants in each group. The distributions of gender and age are similar across the three groups: *mean* $\pm$ *std.* age in *EG* is 13.63 ± 0.92 , *mean* $\pm$ *std.* age in *CG1* is 13.88 ± 0.99 , and *mean* $\pm$ *std.* age in *CG2* is 13.75 ± 1.49 . To achieve balance, factors other than the study variable (i.e., the intervention received) should remain consistent between the control and experimental groups throughout the study.
- **Intervention.** During each session, a *CeM*-based intervention is administered to the adolescent in *EG*. Meanwhile, the adolescents in *CG1* receive a *ZenVR*-based intervention, and the adolescents in *CG2* receive a *mind*-based intervention. In addition, after each treatment session, we ask the adolescents to complete the GAD-7, EMA, and MDQ scales. These data are combined for the subsequent qualitative analysis of intervention effects.
- **Post-test.** After 8 intervention sessions over 4 weeks, all participants are tested in the same manner as the pre-test to obtain their scores on the HAMA-14 scale. These scores are compared with the pre-test scores through between-group and within-group analyses to quantitatively evaluate the therapeutic effect of *CeM*.

5 CONCLUSION, LIMITATION, AND FUTURE WORK

In this paper, we propose a cognition-enhanced meditation therapy (*CeM*) for early intervention in adolescent anxiety disorders. Through a comprehensive main user study involving 24 adolescents with anxiety disorders, experimental results demonstrate that *CeM* exhibits significantly higher intervention effects compared to existing VR meditation therapies, and demonstrates superior intervention effects in alleviating the persistent worry, insecurity, and difficulty concentrating associated with adolescent anxiety disorders.

Although *CeM* has promising efficacy in early intervention for adolescent anxiety disorders, it has several limitations. First, current intervention VEs in *CeM* are limited in scale and lack dynamic adaptability to adolescents' real-time psychological fluctuations. Second, the guiding audio in the awareness phase relies on the concatenation of preset clips, hindering automated personalization. Third, the stimulus phase exclusively employs Taichi movements; this movement regimen uniformity restricts the therapy's generalizability and long-term adolescent adherence. In our future work, we will first construct a large generative model to generate highly adaptive VEs that accommodate individualized psychological conditions. Second, we will develop a context-aware meditation

audio generation model based on multimodal clinical data, thereby advancing fully data-driven psychological interventions. Third, we will incorporate a broader spectrum of interactive exercises that synergize dynamic and static elements into our mind-body early intervention framework, further enhancing cognitive abilities and maximizing overall intervention outcomes.

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